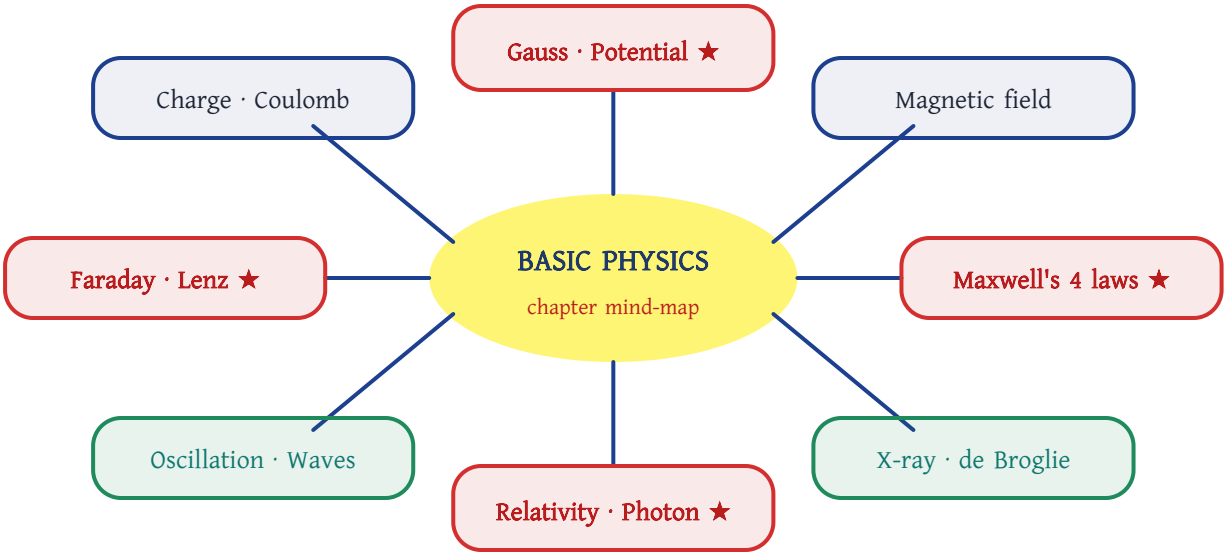
Basic Physics

Electromagnetism · Waves · Relativity · Quantum: exam-ready in 14 sheets



↑ the whole syllabus on one page: red = high-yield, green = formula-driven

High-yield list

① k, ε_0, μ_0, e, h values
② Gauss: $\Phi = q/\varepsilon_0$
③ Maxwell's 4 laws
④ γ, time dilation
⑤ $hf = \varphi + K_{max}$
⑥ $\lambda = h/p$, Compton shift

Syllabus coverage

Charge, **Coulomb's law**, electric field, **Gauss's law**, potential, magnetic field; Faraday's law, **Maxwell's equations** / waves & oscillations / special relativity / EM waves, **photoelectric effect**, quantum theory of light, X-ray & diffraction, Compton effect / de Broglie waves, phase & group velocity, wave function & wave equation.

Legend: **yellow** = key term · **red circle** = must-memorise value · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ Charge, Coulomb, electric field ..... p. 2
✓ Gauss's law, potential ..... p. 3
✓ Magnetic field, Faraday, Lenz ..... p. 4
✓ Maxwell's equations, EM spectrum ... p. 5
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✓ Waves & wave equation ..... p. 7
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✓ Photon, photoelectric effect ..... p. 9
✓ X-rays, Bragg ..... p. 10

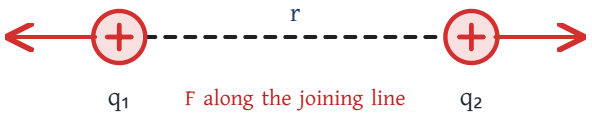
✓ Compton, de Broglie ..... p. 11
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✓ Formula & constants sheet ... p. 13
✓ Practice MCQs + written ..... p. 14

1 Electric Charge & Coulomb's Law

Definition: **Charge (q)**: the property of matter that makes it feel an **electric force**. SI unit: **coulomb (C)** = 1 A·s. Two kinds: + and −.

Like charges repel, unlike attract. Check the sign first!

- **Quantised:** $q = \pm n e$, where $e = \mathbf{1.602 \times 10^{-19}}$ C
- **Conserved:** total charge of an isolated system never changes
- **Invariant:** same in every frame (unlike relativistic mass)
- 1 C ≈ $\mathbf{6.25 \times 10^{18}}$ electrons



like charges push apart

Coulomb's law:

$F = k\, q_1 q_2 / r^2$

$k = 1/4\pi\varepsilon_0 = \mathbf{8.99 \times 10^9}$ N·m²/C²

$\varepsilon_0 = \mathbf{8.854 \times 10^{-12}}$ C²/N·m²

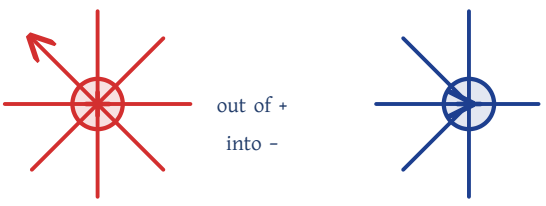
In a medium: $F_{medium} = F_{vacuum} / \varepsilon_r$ → water ($\varepsilon_r \approx 80$) cuts the force **80 times**, which is why salt dissolves.

Exam trap: Double r → F becomes **¼** (inverse square). Double *both* charges → **4F**. Halve r → **4F**.

Worked example ✍️: Two 1 μC charges, 30 cm apart: $F = 9\times10^9 \times (10^{-6})^2 / (0.3)^2 = 9\times10^{-3} / 0.09 =$ **0.1 N** (repulsive)

2 Electric Field (E)

Definition: Force per unit positive test charge: $\mathbf{E = F / q_0}$. A **vector**; unit N/C = V/m. Point charge: $\mathbf{E = k\, q / r^2}$.

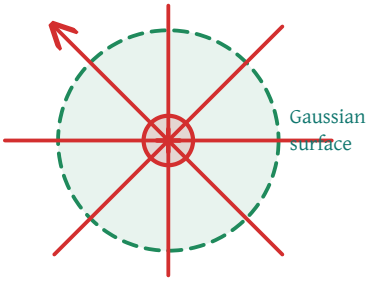


- Lines start on +, end on −
- They **never cross** (E has one direction per point)
- Closer lines = stronger field
- Meet a conductor **at 90°**; E = 0 inside it

Memory trick: Field lines behave like water from a fountain: they flow **out of +** and drain **into −**. Analogy: gravity field g = F/m ↔ electric field E = F/q.

3 Gauss's Law

Statement: Total **electric flux** through any closed surface = enclosed charge / ε_0: $\oint \mathbf{E}\cdot d\mathbf{A} = q_{enc} / \varepsilon_0$



Flux Φ_E = EA cos θ (N·m²/C) = field lines piercing a surface.

Flux does NOT depend on the size or shape of the surface, only on q inside!
Charge **outside** the surface → net flux **0** (every line in also goes out).

Charge distribution	Field E	Remember
Point charge / outside a sphere	kq/r^2	acts like a point charge
Infinite line (λ C/m)	$\lambda / 2\pi\varepsilon_0 r$	∝ 1/r
Infinite plane sheet (σ C/m²)	$\sigma / 2\varepsilon_0$	independent of distance
Just outside a conductor	σ / ε_0	twice the sheet value
Inside a conductor	0	charge sits on the surface

Classic MCQ: Charge q at the centre of a cube → flux through **one face** = $\mathbf{q / 6\varepsilon_0}$ (total q/ε_0 shared equally by 6 faces).

4 Electric Potential (V)

Definition: Work done per unit positive charge to bring it from infinity: **V = W / q**. A **scalar**; unit volt = J/C. Point charge: **V = kq / r** (∝ 1/r, not $1/r^2$)

- **E = -dV/dr**: the field points from **high to low potential**
- Energy: ΔU = qΔV; **1 eV = 1.602 × 10^{-19} J** (energy of e through 1 V)
- **Equipotentials** ⊥ field lines: no work along them; a conductor is one equipotential

	Field E	Potential V
type	vector	scalar
point q	kq/r^2	kq/r
unit	N/C	J/C = V

Memory trick: **E is the downhill slope of V**: a + charge rolls from high V to low V along E.

Worked example ✍️: V at 10 cm from a 1 nC charge = $9\times10^{9} \times 10^{-9} / 0.1$ = **90 V**

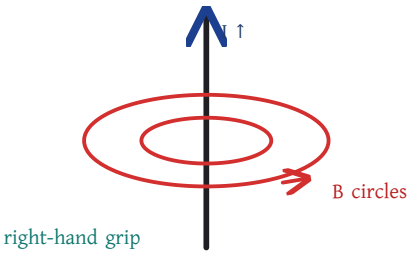
5 Magnetic Field (B)

Definition: Region where a **moving charge or current** feels a force. Unit **tesla (T)** = N/(A·m); **1 T = 10^{4} gauss** · μ_0 = **$4\pi \times 10^{-7}$ T·m/A**

Lorentz force: F = q(**v** × **B**)

|F| = **qvB sin θ** · on a wire: **F = BIL sin θ**

F ⊥ v → **no work** is done; speed stays fixed, the path is a circle with r = mv / qB.



Source	B	Note
Long straight wire	$\mu_0 I / 2\pi r$	∝ 1/r
Centre of a circular loop	$\mu_0 I / 2R$	N turns → ×N
Long solenoid (n turns/m)	$\mu_0 n I$	uniform inside

Memory trick: Fleming's **left** hand = **motor**: **F**irst finger = **F**ield, se**C**ond = **C**urrent, thu**M**b = **M**otion. Right hand = generator.

Exam trap: Charge moving **parallel** to B (θ = 0°) or at rest → magnetic force = **0**.

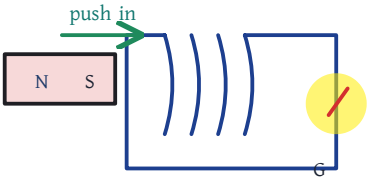
6 Faraday's Law & Lenz's Law

Faraday: A **changing magnetic flux** induces an emf: **$\varepsilon = -N\, d\Phi_B/dt$**, where Φ_B = BA cos θ (weber, Wb = $T\cdot m^2$)

Lenz's law (the minus sign): the induced current **opposes the change** that causes it. This is energy conservation.

Change B, area A or angle θ → emf. Rotating coil = **AC generator**: ε = NBAω sin ωt

Moving rod: **ε = BLv** (motional emf)



needle kicks only while moving

Uses: generator, **transformer** ($V_s/V_p = N_s/N_p$, **AC only**), induction cooker, card readers, eddy-current brakes.

Worked example ✍️: 100 turns, flux 0.02 Wb → 0 in 0.1 s: |ε| = 100 × 0.02 / 0.1 = **20 V**

7 Maxwell's Equations (1865)

Four equations that **unify electricity, magnetism and light**:

#	Law	Integral form	Meaning
1	Gauss (E)	$\oint E \cdot dA = q / \varepsilon_0$	charges create E
2	Gauss (B)	$\oint B \cdot dA = 0$	**no magnetic monopoles**
3	Faraday	$\oint E \cdot dl = -d\Phi_B/dt$	changing B → E
4	Ampère–Maxwell	$\oint B \cdot dl = \mu_0 I + \mu_0\varepsilon_0\, d\Phi_E/dt$	current & changing E → B

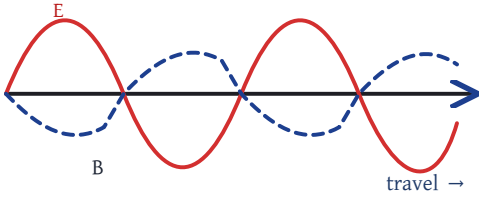
Differential: $\nabla \cdot E = \rho/\varepsilon_0$ · $\nabla \cdot B = 0$ · $\nabla \times E = -\partial B/\partial t$ · $\nabla \times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t$

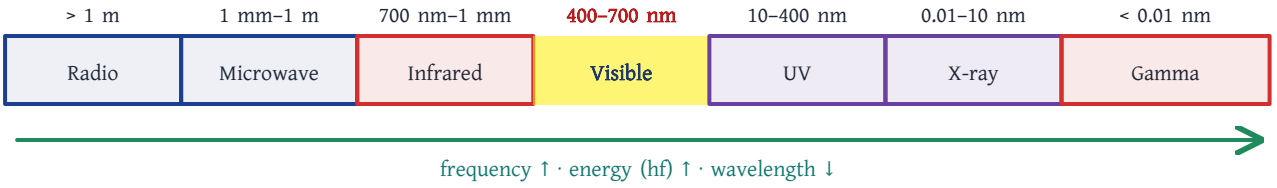
Displacement current: $I_d = \varepsilon_0\, d\Phi_E/dt$: Maxwell's own addition. It lets "current" flow across a **charging capacitor's gap**, and it is what makes EM waves possible.

Memory trick: Order **G-G-F-A**: two **divergences** (sources: charge yes, monopole no) and two **curls** (circulation: Faraday, Ampère).

8 EM Waves & the Spectrum

- Maxwell predicted them; **Hertz (1887–88)** produced them
- Speed **$c = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$ m/s**
- **E ⊥ B ⊥ direction: transverse**, E/B = c
- Need **no medium**; carry energy & momentum





- **Microwave**: Wi-Fi (2.4/5 GHz), mobile, radar, oven
- **Infrared**: TV remote, thermal camera; optical fibre uses near-IR (1310/1550 nm)
- **UV**: sterilisation, forged-note check
- **X-ray**: medical imaging, crystal structure · **Gamma**: radiotherapy

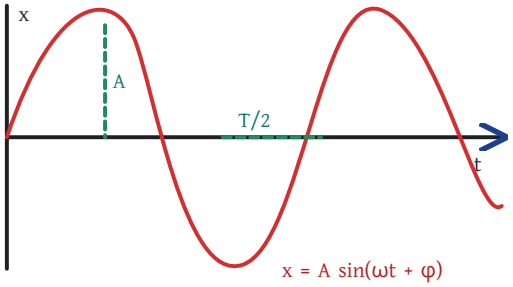
Memory trick: "**R**aging **M**artians **I**nvaded **V**enus **U**sing **X**-ray **G**uns" = increasing frequency.

Exam trap: In vacuum **every** EM wave travels at the same c; they differ only in f and λ. Sound is **not** an EM wave.

9 Oscillations & Simple Harmonic Motion

SHM: Periodic motion in which the **restoring force ∝ displacement** and points back to the mean position: $F = -kx \Rightarrow a = -\omega^2 x$

Term	Meaning
Amplitude A	max displacement
Period T	time of 1 oscillation
Frequency f	f = 1/T (Hz)
Angular freq. ω	ω = 2πf = 2π/T
Phase φ	starting position



Key equations:

$x = A \sin(\omega t + \varphi)$

$v = \omega\sqrt{(A^2 - x^2)}$

$v_{max} = \mathbf{A\omega}$ (at mean position)

$a_{max} = \mathbf{A\omega^2}$ (at the extremes)

Total energy $E = \frac{1}{2}kA^2$ (constant)

KE max at mean, PE max at extremes

Oscillator	Period	Depends on
Spring–mass	$\mathbf{T = 2\pi\sqrt{(m/k)}}$	m, k (not g)
Simple pendulum (small θ)	$\mathbf{T = 2\pi\sqrt{(L/g)}}$	L, g (not mass, not amplitude)
LC circuit (electrical)	$f = 1 / 2\pi\sqrt{(LC)}$	L, C → radio tuning

Worked example ✍️: k = 100 N/m, m = 1 kg → ω = √(k/m) = 10 rad/s, T = 2π/10 ≈ **0.63 s**

Exam trap: Pendulum on the Moon (g/6) → T becomes **√6 ≈ 2.45×**. Double the bob's mass → **no change**. Length ×4 → T ×2.

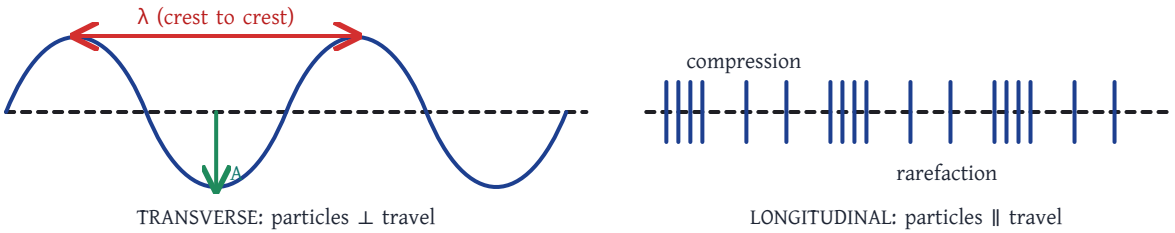
Damped, forced & resonance

- **Damped:** friction drains energy → amplitude decays (shock absorbers)
- **Forced:** an external periodic force drives the system
- **Resonance:** driving frequency = natural frequency → **maximum amplitude** (radio tuning, pushing a swing in step)

Memory trick: **Mean = max speed, Extreme = max acceleration.** Like a swing: fastest at the bottom, stops and turns at the ends.

10 Waves & the Wave Equation

Definition: A wave is a travelling disturbance that **carries energy, not matter**.



TRANSVERSE: particles ⊥ travel

LONGITUDINAL: particles ∥ travel

Type	Examples	Key fact
Mechanical	sound, water, string	need a medium
Electromagnetic	light, radio, X-ray	travel in vacuum
Transverse	string, all EM waves	can be **polarised**
Longitudinal	sound in air	cannot be polarised

Wave formulas:

$v = f\lambda$

$k = 2\pi/\lambda$ (wave number), $v = \omega/k$

$y(x, t) = A \sin(kx - \omega t)$

String: $v = \sqrt{(T/\mu)}$

Wave equation: $\partial^2y/\partial x^2 = (1/v^2)\,\partial^2y/\partial t^2$; any shape $f(x - vt)$ solves it.

- Sound in air ≈ **343 m/s** (20 °C; 332 at 0 °C)
- $v_{solid} > v_{liquid} > v_{gas}$; no sound in vacuum
- Constructive: path difference $n\lambda$; destructive: $(n + ½)\lambda$
- Standing wave: node to node = **$\lambda/2$**
- Beats: $f_{beat} = |f_1 - f_2|$
- Doppler: source approaching → higher pitch

Audible range **20 Hz – 20 kHz**; below = **infrasonic** (earthquakes), above = **ultrasonic** (SONAR, medical scans). Intensity ∝ **A^2**: double the amplitude → 4× the energy.

Exam trap: When a wave enters a new medium its **frequency stays the same**; only speed and λ change.

IT link ✍: Wi-Fi at 2.4 GHz: $\lambda = c/f = 3\times10^8 / 2.4\times10^9$ = **12.5 cm** · FM 100 MHz: λ = **3 m** → that is why antennas are sized in $\lambda/2$ or $\lambda/4$.

11 Special Theory of Relativity

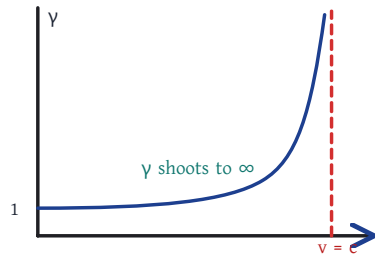
Postulates: ① The laws of physics are the **same in every inertial frame**. ② The speed of light in vacuum, **$c = 3 \times 10^8$ m/s**, is the same for all observers, whatever the source's motion.

Michelson–Morley (1887): no "ether" found, supporting postulate 2

Everything follows from the **Lorentz factor: $\gamma = 1 / \sqrt{(1 - v^2/c^2)}$** (always ≥ 1)

Effect	Formula	Meaning
Time dilation	$\Delta t = \gamma\, \Delta t_0$	moving clock runs slow
Length contraction	$L = L_0 / \gamma$	only along the motion
Relativistic mass	$m = \gamma\, m_0$	→ ∞ as v → c
Mass–energy	**$E = mc^2$**	rest energy m_0c^2
Energy–momentum	$E^2 = (pc)^2 + (m_0c^2)^2$	photon: E = pc

v	0.6c	0.8c	0.866c	0.99c
γ	1.25	1.67	2	7.09



Muon proof ✍️: Lifetime at rest 2.2 μs. At 0.99c: Δt = 7.09 × 2.2 ≈ **15.6 μs**, so muons from the upper atmosphere reach the ground.

$E = mc^2$ ✍️: 1 g of mass → $E = 10^{-3} \times (3\times10^8)^2 = \mathbf{9 \times 10^{13}}$ **J**: the basis of nuclear power.

Exam trap: Nothing with mass reaches c: $u = (u' + v)/(1 + u'v/c^2) \Rightarrow c$ + anything = **c**. t_0, L_0 = **proper** values (rest frame). Charge never changes with speed.

Memory trick: "**Moving clocks run slow, moving rulers shrink**": γ multiplies time, divides length.

IT link: GPS satellite clocks need relativity corrections (≈ 38 μs/day net), or positions drift by km per day.

12 Quantum Theory of Light

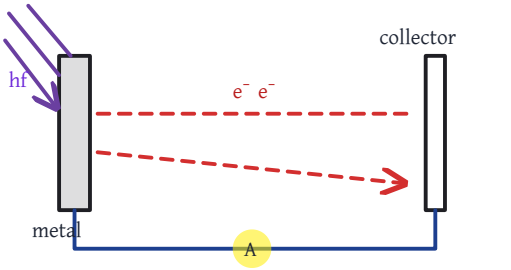
Photon: Light travels as packets (**quanta**) of energy **$E = hf = hc/\lambda$**, momentum **$p = h/\lambda = E/c$**, rest mass **zero**, speed c.

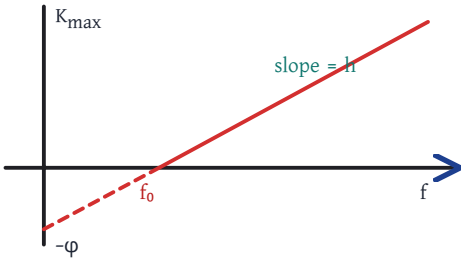
h = **6.626×10^{-34} J·s** · shortcut **hc = 1240 eV·nm** → E(eV) = 1240/λ(nm): red 620 nm = **2 eV**

Wave–particle duality: interference & diffraction → **wave**; photoelectric effect & Compton effect → **particle**.

13 Photoelectric Effect

Definition: Emission of electrons from a metal surface when light of **high enough frequency** falls on it. Observed by Hertz (1887); explained by Einstein (1905, Nobel 1921).





Einstein's equation: $hf = \varphi + K_{max}$ · $K_{max} = eV_0$ (V_0 = stopping potential) · threshold $f_0 = \varphi/h$, $\lambda_0 = hc/\varphi$

You increase...	Photocurrent	K_{max} / V_0
Intensity (f fixed, $f > f_0$)	increases (more photons)	no change
Frequency (intensity fixed)	≈ no change	increases linearly
Anything, but $f < f_0$	no emission, however bright · emission is instant	

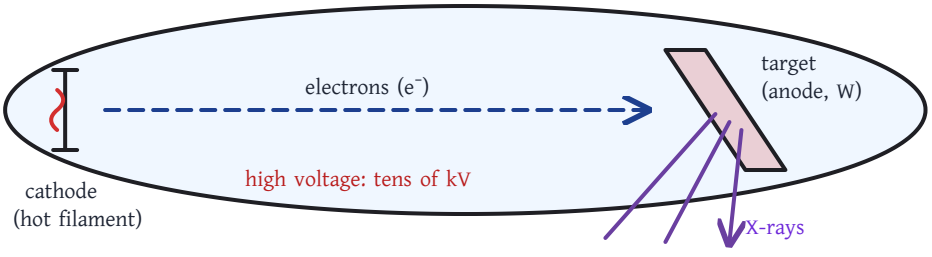
Exam trap: Brighter light means **more** electrons, not **faster** ones. Classical wave theory fails on all three points: threshold, K_{max} vs f, no time lag.

Worked example ✍️: 400 nm light on sodium (φ = 2.3 eV): E = 1240/400 = 3.1 eV → K_{max} = 3.1 − 2.3 = **0.8 eV** → V_0 = 0.8 V

IT link: photodiodes in optical-fibre receivers, CCD/CMOS camera sensors and solar cells all work on photon absorption.

14 X-Rays

Definition: EM waves of **$\lambda \approx 0.01 - 10$ nm**, produced when fast electrons strike a heavy-metal target (usually **tungsten**).



Two parts of the spectrum

- **Continuous** (bremsstrahlung = "braking radiation"): electrons slowing down near nuclei

- **Characteristic lines** (K_α, K_β): inner-shell electron jumps → fingerprint of the target element

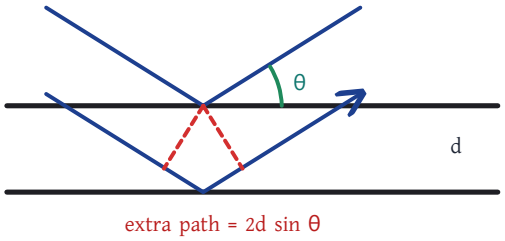
Properties

- Penetrating & ionising; blacken photographic film
- **Not deflected** by E or B fields (uncharged)
- Hard X-rays = short λ = more penetrating

Cut-off wavelength: All of eV goes into one photon: $\lambda_{min} = hc / eV$ → λ_{min}(nm) = 1240 / V(volts). 30 kV tube → λ_{min} ≈ **0.041 nm**

15 X-Ray Diffraction & Bragg's Law

Atomic spacing in crystals (~0.1 nm) ≈ X-ray λ, so a crystal acts as a **3-D diffraction grating** (Laue 1912; W.H. & W.L. Bragg 1913).



Bragg's law:

$2d \sin \theta = n\lambda$ (n = 1, 2, 3 ...)

Uses: finding crystal structure; Rosalind Franklin's X-ray image (1952) revealed the **DNA double helix**.

Exam trap: In Bragg's law θ is the **glancing angle** (ray to the crystal plane), **not** the angle to the normal. The largest possible λ is **2d** (sin θ ≤ 1).

Worked example ✍️: d = 0.28 nm, first order at θ = 15°: λ = 2 × 0.28 × sin 15° = 0.56 × 0.259 ≈ **0.145 nm**

16 Compton Effect

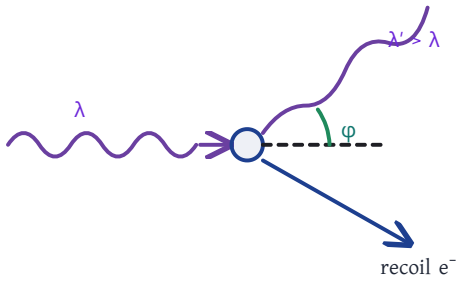
Definition: An X-ray photon scattered by a nearly free electron comes out with a **longer wavelength**. The photon behaves like a billiard ball with momentum p = h/λ.

Compton shift:

$\Delta\lambda = \lambda' - \lambda = (h / m_e c)(1 - \cos \varphi)$

$h / m_e c$ = **2.43 pm** (Compton wavelength)

φ	0°	90°	180°
Δλ	0	2.43 pm	**4.85 pm (max)**



- Shift depends **only on φ**, not on λ or the target material
- It proves photons carry **momentum**: strong evidence for the particle nature of light

Exam trap: Why X-rays, not visible light? Δλ is only a few pm, which is negligible next to 500 nm but large next to 0.07 nm.

17 de Broglie Waves (Matter Waves)

Hypothesis: If light (a wave) acts like particles, then **particles act like waves:** **λ = h / p = h / mv = h / √(2mK)**

Electron through V volts:

λ = **1.226 / √V nm**

100 V → λ ≈ **0.123 nm**, the same scale as X-rays → electrons diffract from crystals!

Why don't we see it? ✍️: Cricket ball, 0.1 kg at 10 m/s: λ = 6.6×10^{-34} / 1 = **6.6×10^{-34} m**, far too small to detect.

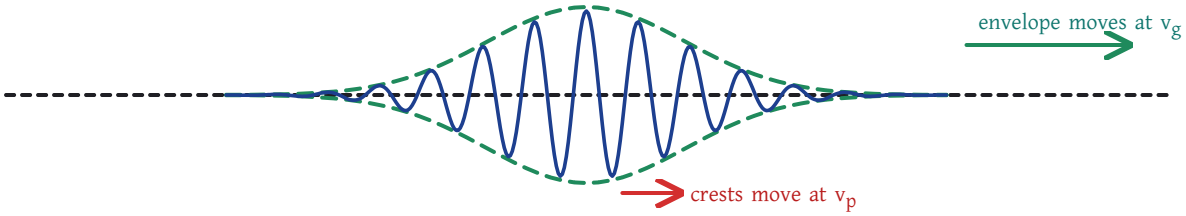
- **Davisson–Germer (1927):** electron diffraction from a nickel crystal (54 V electrons) **confirmed** matter waves
- **Bohr orbits:** 2πr = nλ (whole wavelengths fit) → quantised orbits

Memory trick: **Light** → **particle** (photoelectric, Compton); **particle** → **wave** (de Broglie, Davisson–Germer). Same h links both.

IT link: electron microscopes use tiny electron λ to image chip features far smaller than visible light can resolve.

18 Phase Velocity & Group Velocity

A localised particle = a **wave packet**, i.e. many waves of slightly different k added together.



	Phase velocity v_p	Group velocity v_g
definition	ω / k = E / p	dω / dk = **dE / dp**
matter wave	c^2 / v > c (!)	**= v (particle speed)**
carries	no information	energy & information

Product: **$v_p \cdot v_g = c^2$**

Light in vacuum: $v_p = v_g = c$

Heisenberg: **$\Delta x \cdot \Delta p \geq \hbar/2$**, ħ = h/2π

Exam trap: $v_p > c$ does **not** break relativity, because no signal travels at v_p. Group velocity = dE/dp (some texts misprint dp/dm).

19 Wave Function & Schrödinger Equation

Wave function Ψ(x, t): Holds everything knowable about a particle. Ψ itself is not measurable and may be complex, but **$|\Psi|^2$ = probability density** (Born rule).

- ✓ Normalised: $\int |\Psi|^2 \, dx = 1$
- ✓ Single-valued, finite, continuous
- ✓ $d\Psi/dx$ continuous (where V is finite)
- ✓ Measurement "collapses" Ψ

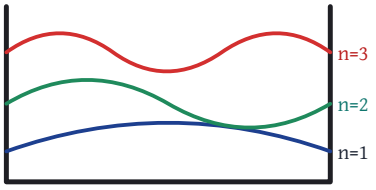
Schrödinger equation:

Time-dependent: $\mathbf{i\hbar \, \partial\Psi/\partial t = -(\hbar^2/2m) \, \partial^2\Psi/\partial x^2 + V\Psi}$

Time-independent (steady states): $-(\hbar^2/2m) \, d^2\psi/dx^2 + V\psi = E\psi$

Particle in a 1-D box (width L): $\mathbf{E_n = n^2h^2 / 8mL^2}$, n = 1, 2, 3 ...

Energy is **quantised**; n = 1 is not zero (zero-point energy). **IT link:** quantum **tunnelling** programs and erases NAND flash cells (SSDs, pen drives).



Night-Before-the-Exam Sheet

Constants: memorise these:

$c = 3 \times 10^8$ m/s

$e = 1.602 \times 10^{-19}$ C

$m_e = 9.11 \times 10^{-31}$ kg

$m_p = 1.673 \times 10^{-27}$ kg

$h = 6.626 \times 10^{-34}$ J·s

$\hbar = 1.055 \times 10^{-34}$ J·s

$k = 8.99 \times 10^9$ N·m^2/C^2

$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2/N·m^2

$\mu_0 = 4\pi \times 10^{-7}$ T·m/A

1 eV $= 1.602 \times 10^{-19}$ J

$hc = 1240$ eV·nm

$h/m_e c = 2.43$ pm

Formula sheet:

$F = kq_1q_2/r^2$ · $E = F/q$

$V = kq/r$ · $E = -dV/dr$

$\oint E \cdot dA = q/\varepsilon_0$ · sheet $\sigma/2\varepsilon_0$

$F = qvB \sin\theta$ · $F = BIL \sin\theta$

$B = \mu_0 I/2\pi r$ · solenoid $\mu_0 nI$

$\varepsilon = -N\, d\Phi/dt$ · $\varepsilon = BLv$

$c = 1/\sqrt{(\mu_0\varepsilon_0)}$ · $v = f\lambda$

$T = 2\pi\sqrt{(m/k)}$ · $2\pi\sqrt{(L/g)}$

$\gamma = 1/\sqrt{(1 - v^2/c^2)}$

$\Delta t = \gamma \Delta t_0$ · $L = L_0/\gamma$ · $E = mc^2$

$E = hf$ · $hf = \varphi + K_{max}$

$\lambda_{min} = hc/eV$ · $2d \sin\theta = n\lambda$

$\Delta\lambda = (h/m_e c)(1 - \cos\varphi)$

$\lambda = h/p$ · e^-: $1.226/\sqrt{V}$ nm

$v_p = \omega/k$ · $v_g = d\omega/dk$

$E_n = n^2h^2/8mL^2$

Top 12 one-liners for MCQs

1. Electric flux unit: N·m^2/C

2. Inside a conductor: E = 0, V constant
3. $\oint B \cdot dA = 0 \rightarrow$ no monopoles
4. Lenz's law = energy conservation
5. Displacement current: Maxwell
6. Pendulum T is independent of mass
7. Sound (longitudinal) cannot be polarised
8. $\gamma(0.6c) = 1.25$, $\gamma(0.8c) = 1.67$
9. K_{max} depends on f, not intensity
10. Compton shift max at 180° = 4.85 pm
11. Davisson–Germer proved matter waves
12. $|\Psi|^2$ = probability density

Top traps: $1/r^2$ (E, F) vs 1/r (V) · θ in Bragg = glancing angle · brighter ≠ faster electrons · $v_p > c$ is allowed · frequency is unchanged when entering a new medium

All tricks together: Fountain (field lines) · E = downhill slope of V · **F**ield-**C**urrent-**M**otion (left hand = motor) · **GGFA** (Maxwell) · **RMIVUXG** (spectrum) · mean = max speed · moving clocks slow, rulers shrink

✓ Practice MCQs (exam pattern, 10 marks)

1. If the distance between two point charges is tripled, the force becomes:
   (a) F/9
   (b) F/3
   (c) 3F
   (d) 9F
2. SI unit of magnetic flux:
   (a) tesla
   (b) henry
   (c) weber
   (d) gauss
3. Which Maxwell equation rules out magnetic monopoles?
   (a) $\oint E \cdot dA = q/\varepsilon_0$
   (b) $\oint E \cdot dl = -d\Phi_B/dt$
   (c) $\oint B \cdot dl = \mu_0 I$
   (d) $\oint B \cdot dA = 0$
4. Wavelength of a 3 GHz microwave:
   (a) 1 cm
   (b) 10 cm
   (c) 1 m
   (d) 10 m
5. A pendulum's length is made 4 times. Its period becomes:
   (a) 2T
   (b) T/2
   (c) 4T
   (d) T

6. Lorentz factor γ at v = 0.8c:

(a) 1.25

(b) 2

(c) 1.67

(d) 0.6

7. Photoelectric effect, $f > f_0$ fixed. Raising the intensity increases:

(a) K_{max}

(b) photocurrent

(c) V_0

(d) f_0

8. Energy of a 620 nm photon:

(a) 1 eV

(b) 3 eV

(c) 6.2 eV

(d) 2 eV

9. Compton shift is maximum at a scattering angle of:

(a) 0°

(b) 45°

(c) 180°

(d) 90°

10. de Broglie λ of an electron accelerated through 100 V:

(a) 0.123 nm

(b) 0.012 nm

(c) 1.23 nm

(d) 12.3 nm

Answer key → 1-a · 2-c · 3-d · 4-b · 5-a · 6-c · 7-b · 8-d · 9-c · 10-a

Written question: State Einstein's photoelectric equation. Light of wavelength 300 nm falls on a metal of work function 2.5 eV. Find (i) K_{max}, (ii) the stopping potential, (iii) the threshold wavelength. 8 marks

1. **Equation:** $hf = \varphi + K_{max}$: photon energy = work function + max kinetic energy of the ejected electron. 2
2. **Photon energy:** $E = hc/\lambda = 1240 / 300 =$ **4.13 eV** 2
3. **(i)** $K_{max} = 4.13 - 2.5 =$ **1.63 eV** $(= 2.6 \times 10^{-19}$ J) 2
4. **(ii)** $eV_0 = K_{max} \rightarrow V_0 =$ **1.63 V** · **(iii)** $\lambda_0 = hc/\varphi = 1240 / 2.5 =$ **496 nm** 2

Before the exam ✓

- ✓ Can write all 4 Maxwell equations + meaning
- ✓ Can draw E-field lines & Gauss surfaces
- ✓ Know γ table & both relativity effects
- ✓ Can solve $hf = \varphi + K_{max}$ in eV units
- ✓ Know Bragg, Compton, de Broglie formulas

> **Also practise (written):**
> ① Derive $E = kq/r^2$ from Gauss's law
> ② Explain displacement current

③ Time dilation with the muon example
④ Compton shift derivation (outline)